# Colour–Number Identities and Challenge-Style Re-Encoding for Player Tracking and Identification in 3x3 Basketball

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**Abstract**

We describe our solution to the ABC 2027 TrackID3x3 Challenge: for every frame of a fixed-camera 3x3 basketball video, output the court position (in cm) and the identity (team and jersey number) of each of the six players. TI-HOTA with $\tau = 100$ cm evaluates the output, and a correct position with a wrong identity scores nothing. Our pipeline has four steps:

1. A YOLO11l detector, fine-tuned in three rounds, finds the players, and BoT-SORT links the boxes over time.
2. A single ResNet18 classifier reads the bib colour and the jersey number of every box as one of twelve (colour, number) identities plus "other". It can do this because each team wears the same colour in every video.
3. The classifier probabilities are averaged along each tracklet. A per-frame Hungarian assignment then gives each of the six roster identities at most one box.
4. The bottom-centre of each box is projected onto the court through the official homography. Outliers are removed with a running median and with a physical speed limit, and gaps are interpolated.

The Challenge videos differ from the Training videos in camera viewpoint, resolution, codec and dynamic range. We therefore re-encode the Training videos to imitate the Challenge videos and use the re-encoded frames both for training and for a realistic validation. On two held-out Training videos processed this way, the final system reaches a TI-HOTA of 0.968, with a median position error of 5 cm. We also report the changes that did not help, and an analysis of the metric: because the evaluation compares exactly six predicted rows with six ground-truth rows per frame, TI-HOTA is governed almost entirely by the localisation accuracy of correctly identified players.

## 1 Introduction

Tracking players and knowing who they are is the basis of most sports analytics: distances, speeds, spacing, line-ups and tactical patterns all require trajectories attached to identities. The TrackID3x3 dataset [1] provides 3x3 basketball videos with player boxes, court positions and identities. The ABC 2027 TrackID3x3 Challenge asks for the complete chain, from video to court trajectories of identified players.

Three properties of the Challenge shape our method.

- **Identity is part of the metric.** A predicted point counts only if both its team and its jersey number are right. Jersey numbers also repeat across the two teams (e.g., both teams have a number 6 and a number 9 in one Challenge game), so the team and the number must be recognised together.
- **The ground truth is a projection of the box.** We found that the official court positions are exactly the homography image of the bottom-centre of the annotated boxes. Position accuracy is therefore box accuracy, measured at the feet.
- **The Challenge videos are out of domain.** They come from a camera at a different position, farther away and looking at the court at a flatter angle. They are also 720p, standard dynamic range and strongly compressed, whereas the Training videos are 4K HDR.

Our contributions are:

1. **A joint colour–number identity classifier.** We observed that each of the four teams wears a fixed bib colour throughout the dataset. Team assignment and jersey recognition therefore merge into one 13-way classification of each box: twelve (colour, number) identities plus "other". The classifier is trained only on the Training split.

2. **A Challenge-style re-encoding of the Training videos.** We map the Training videos from HLG HDR to SDR, downscale them to 720p and encode them with H.264 at 3 Mbit/s. The re-encoded videos serve as training data for the detector and the classifier, and as a realistic validation set.

3. **Robust trajectory cleaning.** A running-median despike is combined with a segment filter based on physical speed limits. The filter removes short wrong-player stretches and mid-air positions during jump shots, an error that the oblique Challenge camera amplifies.

4. **A systematic evaluation.** We measure every design decision with TI-HOTA on held-out videos, include the negative results, and analyse which errors the metric actually penalises.

## 2 Related Work

**Multi-object tracking in sports.** Tracking-by-detection methods associate detector outputs over time. SORT [3] and DeepSORT [4] combine a Kalman filter with motion and appearance cues. ByteTrack [5] also uses low-confidence detections, and BoT-SORT [6] adds camera-motion compensation and appearance (ReID) features. Sports scenes are harder than pedestrian scenes because players look alike, move quickly and occlude each other often, as shown by SoccerNet-Tracking [7] and SportsMOT [8]. TrackID3x3 [1] adds identification and court localisation, and introduces TI-HOTA, an identity-aware variant of HOTA [2] in court coordinates. Its baseline uses ID-switch merging and reaches a TI-HOTA of 85.5% on the indoor subset and 71.0% on the outdoor subset (their Table 3, with $\tau = 1$ m).

**Player identification.** Jersey numbers are usually read by dedicated digit recognisers or OCR [9], and teams are usually separated by clustering shirt colours. The Challenge tutorial baseline follows this pattern: YOLO11x with ByteTrack, EasyOCR for numbers and HSV $k$-means for teams. Our setting allows a simpler and more reliable formulation, because the bib colour of a team is constant and the roster of each game is given. A single classifier then outputs the full identity, and the roster restricts it to six candidates.

## 3 Data and Task Analysis

**Training data.** There are eight Training videos (IMG_0104–0111) recorded with a fixed iPhone at 3840×2160 and 30 fps, about 6–7 minutes each. They total 93,581 frames and 561,256 annotated player boxes. For every frame and each of the six players, the annotation gives a box, a court position in cm and an identity. All videos share one court registration of four court corners. The players form four teams that meet in different pairings. Table 1 lists the properties that matter for our design.

**How the ground truth is built.** We projected the bottom-centre of every annotated box with the homography estimated from the four registered corners [16]. The result reproduces the official court positions to within 0.001 cm on all videos that have them. The positions are therefore not measured separately: they inherit any error of the box bottom. Two consequences follow:

- We must use the official registration file, even where the projected court lines visibly deviate from the painted lines.

- The detector must place the bottom edge of the box the way the annotators did. Detector fine-tuning matters more than any change to the tracker.

**Table 1:** Training and Challenge videos. The two splits differ in viewpoint, resolution, dynamic range and compression.

| | Training (8 videos) | Challenge (2 videos) |
|---|---|---|
| Camera | fixed, elevated, diagonal view | fixed, other position, farther, flatter angle |
| Resolution | 3840×2160, 30 fps | 1280×720, 30 fps |
| Codec | HEVC Main10, HLG HDR (BT.2020), 29 Mbit/s | H.264 Main, SDR (BT.709), 3 Mbit/s |
| Frames | 10,706–13,228 per video | 11,536 and 11,823 |
| Player height (720p) | ≈ 85–150 px (4K: 250–450 px) | median ≈ 90 px |
| Roster | tracks, teams, numbers, game start, offence | teams and numbers only |

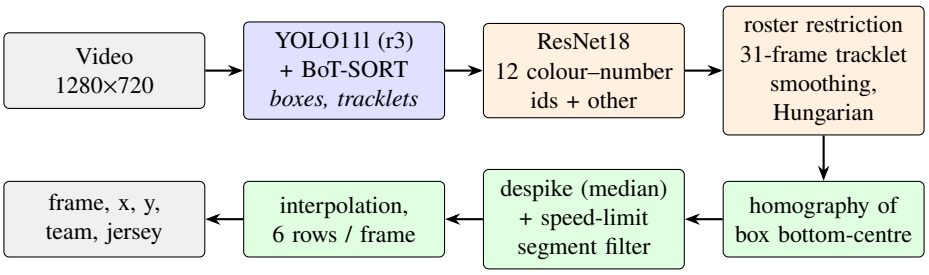



**Figure 1:** Overview. Blue: detection and tracking (Sec. 4.2). Orange: identification and assignment (Sec. 4.3). Green: localisation and cleaning (Sec. 4.4).

**Cleaning.** We drop one off-roster box (a spectator in IMG_0110). In IMG_0105, 25 boxes are duplicated; for each duplicate we keep the box whose foot point continues the track. We also read all videos sequentially: seeking with OpenCV can be off by up to three frames on these files, and a one-frame offset already costs 0.01 TI-HOTA.

**Fixed bib colours.** We inspected all Training videos and found that each team always wears the same bib colour: purple $\{4, 10, 12\}$, yellow $\{6, 9, 10\}$, green $\{6, 9, 11\}$ and pink $\{1, 3, 5\}$. The two Challenge games also use these teams:

- IMG_0114: A = yellow, B = green, with numbers 6 and 9 on both teams.
- IMG_0115: A = purple, B = pink.

Other people in the scene, such as substitutes, referees and spectators, wear other colours.

**Split.** We hold out IMG_0108 and IMG_0109, whose team pairings are the same as in the two Challenge games. We train on the other six videos. The Challenge videos are used only for inference and for label-free sanity checks, never for training, pseudo-labels or manual corrections.

# 4 Method

Figure 1 gives an overview of the pipeline.

## 4.1 Challenge-style re-encoding

The Training videos are HLG-encoded HDR, whereas the Challenge videos are SDR exports. Simply downscaling the Training frames therefore produces images with a different tone, saturation and noise than the Challenge frames. We re-encode each Training video with FFmpeg in four steps:

1. Linearise the HLG signal (nominal peak 203 cd/m$^2$).
2. Convert BT.2020 to BT.709 primaries and apply Hable tone mapping.

**Table 2:** Detector fine-tuning rounds. The images are 720p frames from the six training videos. All rounds use batch 5, image size 1280 and horizontal flips. mAP is measured on the validation frames of each round (clean for r1/r2, Challenge-style for r3).

| Round | Init | Images | Optimiser, epochs | Augmentation | mAP50 / 50–95 |
|---|---|---|---|---|---|
| r1 | COCO YOLO11l | 7.1k clean | AdamW $2{\cdot}10^{-3}$, cos, 10 (best 2) | mosaic, scale 0.6 | 0.969 / 0.792 |
| r2 | r1 best | 14.1k clean | SGD $2{\cdot}10^{-3}$, 8 | mosaic 0.5, scale 0.3 | 0.995 / 0.668 |
| r3 | r2 last | 14.1k re-encoded | SGD $2{\cdot}10^{-3}$, 6 | + rot. 5°, shear 3°, persp. | 0.994 / 0.666 |

3. Downscale to 1280×720 by area averaging.

4. Encode with H.264 Main at 2990 kbit/s, matching the Challenge bit rate.

We keep the frame timing unchanged, so every frame keeps its index and its labels; the 4K box coordinates are simply divided by three. We use the re-encoded videos in three places: as extra training images for the detector (round 3), as extra training crops for the identity classifier, and as our main validation set.

### 4.2 Detection and tracking

We fine-tune Ultralytics YOLO11l [10], initialised from COCO [11], as a one-class player detector at an input size of 1280. The training images are every fifth frame of the six training videos (14,149 images); validation uses every fifth frame of IMG_0108/0109 (4,572 images). Table 2 summarises the three training rounds.

- **Round 1** uses every second training image, AdamW and a cosine schedule. Early stopping selected epoch 2.
- **Round 2** continues from round 1 on all images with SGD and lighter augmentation.
- **Round 3** continues from round 2 on the Challenge-style frames. It adds rotation (±5°), shear (±3°) and perspective augmentation to soften the change of viewpoint.

We select checkpoints by TI-HOTA of the full pipeline, not by detection mAP, for the reason given in Sec. 5.

At inference we keep detections above confidence 0.1 and track them with BoT-SORT [6]. The tracker settings are: high/low/new-track thresholds 0.25/0.1/0.25, a 90-frame buffer and match threshold 0.8. We switch off camera-motion compensation because the camera is fixed, and we switch off ReID (see Sec. 5.1). The tracker only groups boxes into tracklets; identities come from the next stage.

### 4.3 Identity classification and assignment

**Classifier.** Each box is enlarged by 10% on every side and resized to 80×160 pixels. A ResNet18 [12] initialised on ImageNet [13] predicts 13 classes: the 12 (colour, number) identities and "other". Training crops come from the ground-truth boxes of the six training videos, cut from both the clean frames and the Challenge-style frames.

To obtain "other" examples, we run a COCO person detector on the Training frames. Detections with IoU < 0.1 to every player box become negatives (about 33k for training and 10k for validation).

The model is trained for 4,800 steps with batch 128, class-balanced sampling, label smoothing 0.05, AdamW [14] at $10^{-3}$ and a one-cycle schedule. Augmentation simulates the Challenge view:

- box jitter and down/up-scaling to as little as 36 px height;
- rotation of ±6°;
- brightness, saturation and gamma changes;
- JPEG artefacts.

We use no horizontal flip, because it mirrors the digits, and no hue shift, because hue carries the team. On Challenge-style validation crops, restricted to the six roster identities of each video (as the pipeline uses them), the classifier is correct on 98.8% of crops.

**Assignment.** For a video with roster identities $\mathcal{I} = \{1, \ldots, 6\}$, we keep the six corresponding probabilities of every box and fold all others into "other". Single crops are noisy because of occlusion, motion blur and turned backs, so we average the probability vectors along each tracklet with a centred 31-frame moving window. Unlike a vote over the whole tracklet, a moving window follows an identity switch inside a tracklet. In every frame $t$ we then solve

$$\max_{\pi} \sum_{(b,k)\in\pi} \bar{p}_{b,k} \tag{1}$$

over one-to-one matchings $\pi$ between the boxes $b$ and the identities $k \in \mathcal{I}$, using the Hungarian algorithm [15]. A matched pair is kept only if $\bar{p}_{b,k} \geq 0.2$ and $\bar{p}_{b,k} > 0.5\,\bar{p}_{b,\text{other}}$. The assignment guarantees at most one box per identity per frame. Because it works per frame, a fragmented or switched tracklet costs only the frames in which the classifier is wrong.

### 4.4 Localisation, cleaning and interpolation

The court position of an assigned box is the homography image of its bottom-centre (Sec. 3). For each identity we then remove outliers in two steps.

1. **Despike.** A position is dropped if it lies more than 60 cm from the median of the identity's 15 neighbouring observations. This removes isolated wrong assignments.

2. **Segment filter.** The median cannot remove a wrong stretch that is as long as half its window. We therefore cut the trajectory wherever consecutive observations at frames $f_i < f_j$ are farther apart than $50 + 40\,(f_j - f_i)$ cm, that is, faster than about 12 m/s. A piece shorter than 30 observations is dropped if the pieces before and after it can be joined without it, or if it is at an end of the trajectory next to a longer piece.

The segment filter catches two kinds of error: a short assignment to the wrong player, and a jump shot. During a jump the feet leave the floor, so the bottom of the box rises and projects metres away on an oblique camera (Fig. 2).

Finally, we fill the gaps by linear interpolation in time and hold the first and last positions constant, so every frame $1..N$ has exactly six rows. The ground truth also has six rows per frame, including the frames before the game starts.

## 5 Experiments

**Protocol.** We report TI-HOTA with $\tau = 100$ cm and use our own re-implementation, which we checked line by line against the organisers' notebook. For each frame the metric computes the similarity $s = \exp\big(\ln(0.05)\, d^2/\tau^2\big)$ for each ground-truth/prediction pair with the same identity, and $s = 0$ otherwise. It then solves a Hungarian matching and computes $\sqrt{\text{DetA} \cdot \text{AssA}}$ at each threshold $\alpha \in \{0.05, \ldots, 0.95\}$, and averages over $\alpha$. At $\alpha = 0.95$, a match must lie within 13 cm.

We score the mean over IMG_0108 and IMG_0109 in two versions: *clean*, the 4K frames downscaled to 720p, and *Challenge-style*, the re-encoded videos of Sec. 4.1. The Challenge-style version is our main criterion.

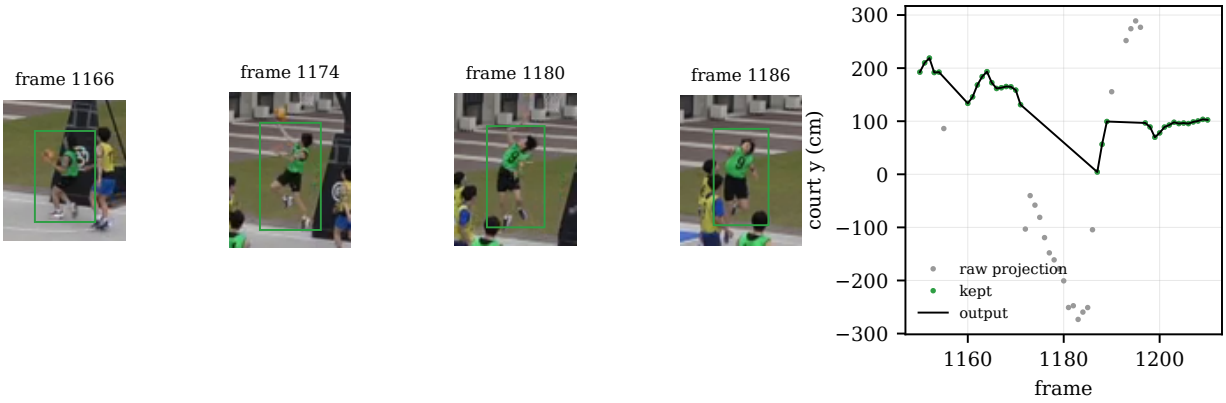



**Figure 2:** A jump shot in Challenge video IMG_0114 (player B9). Left: the feet leave the floor and the box bottom rises. Right: the raw projected court $y$ jumps by more than a metre. The segment filter drops the airborne stretch (grey), and the output (black) interpolates between take-off and landing.

**Table 3:** Development of the pipeline: TI-HOTA, mean of IMG_0108/0109 (never used for training). Every row adds one change to the row above. "–" = not run in that version.

| Configuration | Clean | Challenge-style |
|---|---|---|
| COCO YOLO11l + ByteTrack, identity by whole-tracklet vote (IMG_0108 only) | 0.663 | – |
| + 31-frame smoothing and per-frame Hungarian assignment | 0.887 | – |
| + fine-tuned detector r1, best checkpoint (epoch 2) | 0.932 | – |
| r1, last checkpoint (epoch 10) | 0.962 | – |
| + despike (60 cm from 15-point median) | 0.963 | – |
| + detector r2 | 0.967 | 0.962 |
| + classifier trained with Challenge-style crops | 0.966 | 0.962 |
| + BoT-SORT instead of ByteTrack | – | 0.962 |
| + detector r3 (Challenge-style frames, geometric augmentation) | – | **0.968** |
| + segment filter (final) | – | **0.968** |

**What TI-HOTA rewards.** Both the ground truth and our output contain exactly six rows per frame. Hence at every $\alpha$ we have FP = FN = $6N - \mathrm{TP}$. If $q_\alpha$ is the fraction of rows matched at threshold $\alpha$, then

$$\mathrm{DetA}_\alpha = \frac{q_\alpha}{2 - q_\alpha}. \tag{2}$$

With correct identities, $\mathrm{AssA}_\alpha \approx \mathrm{DetA}_\alpha$. Averaging over $\alpha$ gives TI-HOTA $\approx p/(2-p)$, where $p$ is the mean similarity per row. A loss of 1% in $p$ therefore costs about 2% in TI-HOTA.

Two consequences shape our design:

- Emitting a guessed (interpolated) position is never worse than emitting nothing, so we always output six rows.
- A wrong identity scores zero, whatever its position.

We measured the metric's sensitivity directly. Random position noise of about 13 cm gives 0.95, and noise of about 50 cm gives 0.46. Shifting all frame indices by one frame costs 0.01, and by three frames costs 0.11.

**Main results.** Table 3 traces the pipeline from a COCO detector to the final system.

- **Identity assignment matters most early on.** Replacing a whole-tracklet vote with a smoothed per-frame assignment adds 0.22, because one wrong identity on a long tracklet otherwise removes the whole trajectory.

**Table 4:** Error breakdown of the final system on the Challenge-style validation videos: share of rows and of the similarity lost, per distance band.

| | <15 cm | 15–50 cm | 50–200 cm | >200 cm / wrong id | mean similarity $p$ |
|---|---|---|---|---|---|
| IMG_0108 rows (lost) | 93.1% (0.001) | 6.1% (0.009) | 0.8% (0.006) | 0.0% (0.000) | 0.984 |
| IMG_0109 rows (lost) | 92.7% (0.001) | 6.4% (0.010) | 0.9% (0.007) | 0.0% (0.000) | 0.983 |

- **Detector fine-tuning brings the next largest gain.** It makes the boxes tight at the feet.
- **Mild compression costs little.** The Challenge-style re-encoding lowers the score by only 0.005, so compression and tone mapping are not the main risk.
- **Round 3 adds 0.006 on the Challenge-style videos,** and improves both videos (IMG_0108 0.964 → 0.969, IMG_0109 0.960 → 0.967).
- **The segment filter leaves the validation score unchanged,** because the held-out videos contain no stretch it removes. On the Challenge videos it removes exactly two stretches, and on inspection both are errors: a wrong-player stretch and a jump shot (Fig. 2).

**Detection mAP does not predict TI-HOTA.** The best round-1 checkpoint by mAP50–95 (0.792) scores 0.932 TI-HOTA. The last round-1 checkpoint has a lower mAP50–95 (0.726) but scores 0.962. Later rounds lower mAP50–95 further while raising TI-HOTA. The annotated boxes appear to be precise at the feet, which is what TI-HOTA measures, but looser at the head and arms, which mAP50–95 also weighs. For this reason we select checkpoints only by TI-HOTA of the full pipeline.

**Where the remaining error is.** With the final system, the median position error is 4–5 cm for every player on both videos. Table 4 splits the rows by error:

- 93% of rows are within 15 cm. They earn almost the full score.
- 6% of rows are between 15 and 50 cm. They cause most of the lost similarity, mainly during occlusions, when the visible box bottom is not the feet.
- Less than 1% of rows are off by more than 50 cm.
- No row is missing or assigned to the wrong player.

Identification is therefore essentially solved on the held-out videos, and the remaining gap is localisation under occlusion.

## 5.1 What did not help

Table 5 lists the variants we rejected. We report them because several of them would be reasonable defaults in other tracking problems.

- **Test-time augmentation.** Flips and multiple scales blur the bottom edge of the box, and the share of rows in the 15–50 cm band grows from 7% to 10–12%.
- **BoT-SORT with ReID.** It is 2.5× slower and slightly worse. Identities come from the classifier, so better tracklet appearance matching has little to add.
- **Interpolating over occlusions.** The detector's estimate of an occluded player's feet is still better than interpolating over the occlusion.
- **Temporal smoothing of positions.** It lowered TI-HOTA, because any lag costs points at the strict thresholds.

**Table 5:** Rejected variants (TI-HOTA, IMG_0108/0109; the reference uses the same validation version).

| Variant | Score | Reference |
|---|---|---|
| Test-time augmentation (flip + scales) | 0.956 | 0.962 |
| BoT-SORT with ReID features | 0.960 | 0.962 |
| Replace occluded positions by interpolation (clean) | 0.917–0.962 | 0.963 |
| Segment filter with minimum length 60 | 0.9675 | 0.9680 |
| Detector r3 after 1 / 3 of 6 epochs | 0.953 / 0.966 | 0.968 |

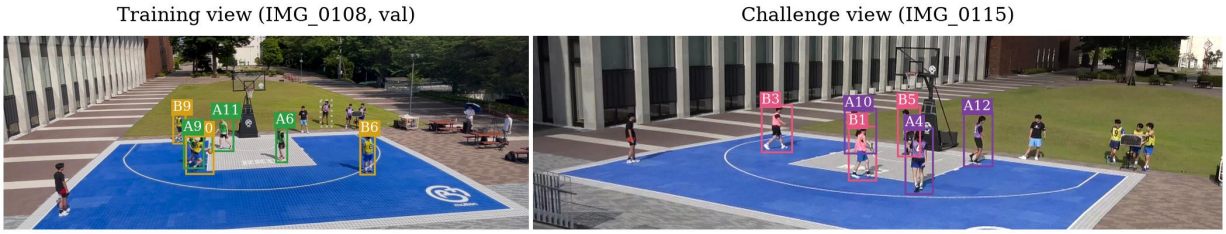



**Figure 3:** Training view (left, held-out IMG_0108 after Challenge-style re-encoding) and Challenge view (right, IMG_0115), with the identities assigned by our pipeline. Dotted grey boxes are detections classified as "other".

- **A longer segment window.** Raising the minimum piece length from 30 to 60 starts removing correct stretches.

### 5.2 The Challenge videos

We have no labels for the Challenge videos, so we checked the outputs in label-free ways.

**Basic checks.** We verified that:

- the official court registration aligns with the painted court lines;
- every frame has the six roster identities;
- drawn frames show the expected bib colours for teams A and B (Fig. 3);
- the trajectories look plausible (Fig. 4).

**Unsupervised statistics can mislead.** We also tracked unsupervised statistics: the number of player tracklets, the share of uncertain crops, and the number of jumps above 1 m per frame. They are useful to spot gross failures, but they did not rank models correctly. With the round-3 detector, the Challenge outputs show more uncertain crops (IMG_0114: 21% → 28%) and more jumps above 1 m (6 → 12) than with round 2. Yet the same shift on the held-out videos (uncertain crops 2.5% → 6.5%, jumps 3 → 6 on IMG_0108) came with a higher TI-HOTA. The new detector simply finds more people, including bystanders, and the classifier rejects them as "other". The round-2 and round-3 Challenge outputs agree within 30 cm on 96% of rows, with a median difference of 8–9 cm.

## 6 Discussion and Limitations

- **Viewpoint.** The change of viewpoint is the risk we cannot measure: the held-out videos share the Training camera. We softened it with geometric augmentation and Challenge-style frames, and with a classifier that sees players from several angles within the Training videos. The flatter viewing angle of the Challenge camera also changes the geometry of the error. A given vertical offset of the box bottom, from a jump or an occlusion, maps to a larger court distance than in the Training view, so we expect the Challenge score to be lower than on validation.

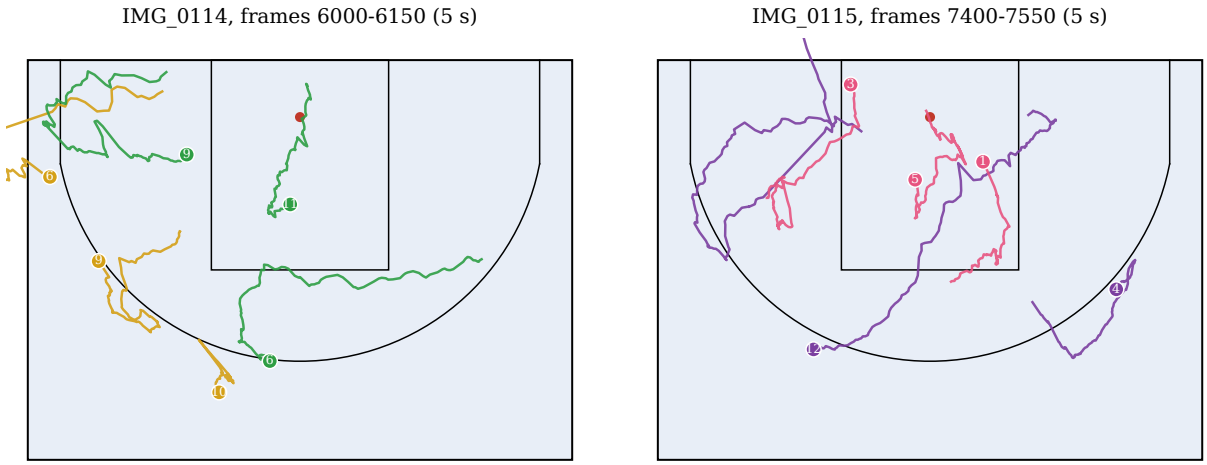



**Figure 4:** Five seconds of predicted trajectories on the two Challenge videos, coloured by bib colour; the dot marks the last position, labelled with the jersey number. The jitter of a few tens of centimetres comes from frame-to-frame variation of the box bottom; we do not smooth positions (Sec. 5.1).

- **Optimistic validation.** We chose all hyper-parameters (the 31-frame window, the 60 cm despike threshold, the checkpoints) on IMG_0108/0109, so the validation numbers are somewhat optimistic.
- **Fixed bib colours.** The joint colour–number formulation relies on fixed team colours. It transfers to leagues with fixed kits, but a new team requires new training crops. Separate colour and number heads would generalise better.
- **Future work.** Most of the remaining error comes from occlusion. Modelling the feet explicitly, for example with pose keypoints or a learned foot point, could reduce it.

## 7 Conclusion

We presented a tracking-and-identification pipeline for 3x3 basketball built around three observations. The court positions are projected box bottoms. Teams wear fixed bib colours, so identity is a 13-way classification. The Challenge videos differ from the Training videos in encoding as well as viewpoint. A fine-tuned YOLO11l detector, a joint colour–number classifier with per-frame Hungarian assignment, and speed-limit trajectory cleaning reach a TI-HOTA of 0.968 on held-out videos re-encoded like the Challenge videos. Every component was trained on the official Training split only. The code, weights and a one-command script that reproduces the submitted predictions accompany this paper.

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